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**COURSE: CSA0910 PROGRAMMING IN JAVA**

## 1. Multiple catch statements JAVA CODE

### PROGRAM:

```
public class MultipleCatch {  
    public static void main(String[] args) {  
        try {  
            int a = 10 / 0;  
            int arr[] = {1, 2, 3};  
            System.out.println(arr[5]);  
        }  
        catch (ArithmeticException e) {  
            System.out.println("Arithmetic Exception: " + e.getMessage());  
        }  
        catch (ArrayIndexOutOfBoundsException e) {  
            System.out.println("Array Index Exception: " + e.getMessage());  
        }  
        catch (Exception e) {  
            System.out.println("General Exception: " + e.getMessage());  
        }  
    }  
}
```

### OUTPUT:

| MultipleCatch.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Output                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <pre>1- public class MultipleCatch {<br/>2-     public static void main(String[] args) {<br/>3-         try {<br/>4-             int a = 10 / 0;<br/>5-             int arr[] = {1, 2, 3};<br/>6-             System.out.println(arr[5]);<br/>7-         }<br/>8-         catch (ArithmeticException e) {<br/>9-             System.out.println("Arithmetic Exception: " + e<br/>10-                .getMessage());<br/>11-         }<br/>12-         catch (ArrayIndexOutOfBoundsException e) {<br/>13-             System.out.println("Array Index Exception: " + e<br/>14-                .getMessage());<br/>15-         }<br/>16-         catch (Exception e) {<br/>17-             System.out.println("General Exception: " + e<br/>18-                .getMessage());<br/>19-         }<br/>20-     }<br/>21- }</pre> | <pre>Arithmetic Exception: / by zero<br/><br/>=== Code Execution Successful ===</pre> |

## 2. StringIndexOutOfBoundsException

### PROGRAM:

```
public class StringIndexExample {  
    public static void main(String[] args) {  
        try {  
            String str = "Hello";  
            System.out.println(str.charAt(10));  
        }  
        catch (StringIndexOutOfBoundsException e) {  
            System.out.println("String Index Out of Bounds Exception");  
        }  
    }  
}
```

### OUTPUT:

| StringIndexExample.java                                                                                                                                                                                                                                                                                                                                                                                                       |    Share | Run | Output                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------|
| <pre>1 public class StringIndexExample {<br/>2     public static void main(String[] args) {<br/>3         try {<br/>4             String str = "Hello";<br/>5             System.out.println(str.charAt(10));<br/>6         }<br/>7         catch (StringIndexOutOfBoundsException e) {<br/>8             System.out.println("String Index Out of Bounds<br/>9                 Exception");<br/>10        }<br/>11    }</pre> |                                                                                                                                                                                                                                                                   |     | <pre>String Index Out of Bounds Exception<br/><br/>=== Code Execution Successful ===</pre> |

### 3. Finally

#### PROGRAM:

```
public class FinallyExample {  
    public static void main(String[] args) {  
        try {  
            int a = 10 / 0;  
        }  
        catch (ArithmeticException e) {  
            System.out.println("Exception: Cannot divide by zero");  
        }  
        finally {  
            System.out.println("Finally block is always executed");  
        }  
    }  
}
```

#### OUTPUT:

| FinallyExample.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Output                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <pre>1 public class FinallyExample {<br/>2     public static void main(String[] args) {<br/>3         try {<br/>4             int a = 10 / 0;<br/>5         }<br/>6         catch (ArithmeticException e) {<br/>7             System.out.println("Exception: Cannot divide by<br/>8                 zero");<br/>9         }<br/>10        finally {<br/>11            System.out.println("Finally block is always<br/>12                executed");<br/>13        }<br/>14    }<br/>15 }</pre> | <pre>Exception: Cannot divide by zero<br/>Finally block is always executed<br/><br/>=== Code Execution Successful ===</pre> |

## 4. Throw

### PROGRAM:

```
import java.util.Scanner;

public class VotingEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        try {
            if (age < 18)
                throw new Exception("Not eligible to vote");
            else
                System.out.println("Eligible to vote");
        }
        catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

### OUTPUT:

| VotingEligibility.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Run | Output                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------|
| <pre>1 import java.util.Scanner; 2 3 public class VotingEligibility { 4     public static void main(String[] args) { 5         Scanner sc = new Scanner(System.in); 6 7         System.out.print("Enter your age: "); 8         int age = sc.nextInt(); 9 10        try { 11            if (age &lt; 18) 12                throw new Exception("Not eligible to vote"); 13            else 14                System.out.println("Eligible to vote"); 15        } 16        catch (Exception e) { 17            System.out.println(e.getMessage()); 18        } 19    } 20 }</pre> |     | <pre>Enter your age: 17 Not eligible to vote  === Code Execution Successful ===</pre> |

## 5. Throws

### PROGRAM:

```
import java.util.Scanner;

public class Voting {

    static void checkAge(int age) throws Exception {
        if (age < 18)
            throw new Exception("Not eligible to vote");
        else
            System.out.println("Eligible to vote");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        try {
            checkAge(age);
        }
        catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

### OUTPUT:

| Voting.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Run            | Output                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------|
| <pre>1- import java.util.Scanner; 2 3- public class Voting { 4 5-     static void checkAge(int age) throws Exception { 6         if (age &lt; 18) 7             throw new Exception("Not eligible to vote"); 8         else 9             System.out.println("Eligible to vote"); 10    } 11 12-    public static void main(String[] args) { 13        Scanner sc = new Scanner(System.in); 14 15        System.out.print("Enter age: "); 16        int age = sc.nextInt(); 17 18-        try { 19            checkAge(age); 20        } 21-        catch (Exception e) { 22            System.out.println(e.getMessage()); 23        } 24    } 25 }</pre> | <div>Run</div> | <pre>Enter age: 24 Eligible to vote  === Code Execution Successful ===</pre> |